

Joanne Hoar

Technology Educator, Software Developer, Game Developer

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Summary

Canadian Software Developer with 20 years of experience in video game programming, scientific modelling and cross-platform development. Expertise in multiple programming languages, game engines and SDKs. Proven track record in leading teams, mentoring developers, impressing customers and building corporate culture. Strong background in C++, Unreal Engine, iOS and Android using agile methodologies and CI/CD. Passion for algorithmic correctness, mathematics, artificial intelligence and nature.

Experience

Technology Instructor

Jan 2025 – present

Robertson College (contract)

C# / .NET / ASP .NET Core / Java / JavaScript / React / Git / GitHub / Azure / SQL

Instructed 20+ course offerings across Full Stack Development and Data Analytics diploma programs. Delivered live coding sessions, maintained open-source course materials on GitHub, and achieved an average student satisfaction rating of 4.5/5.0.

Senior Software Developer

2022 – 2025

bitHeads Inc.

Unreal Engine / C++ / Objective-C / Swift / iOS / Android Native / Java / Kotlin / PlayStation / Xbox / Nintendo Switch / Node.js / Groovy / C# / Shell Script

Developed and maintained cross-platform client SDKs for a backend-as-a-service platform used by 50+ published games across 12 platforms including PlayStation, Xbox, Nintendo Switch, iOS, and Android. Engineered CI/CD and TDD pipelines and managed customer relationships with AAA and Indie studios.

Co-founder/ Lead Developer

2011 – 2022

Silicon Hanna Inc. (self-employed)

Unreal Engine / Objective-C / Swift / iOS / Android Native / Kotlin / Augmented Reality / MacOS / Windows

Designed, coded and published eleven mobile games for iOS and Android, from concept and game design through to commercial app-store release. Experimented with cutting-edge device technologies including ARKit and other emerging SDKs.

Tools Programmer

2008 – 2011

Fekete Associates Inc. (S&P Global)

C++ / MFC / Windows / Perforce / Incredibuild

Developed tools that used mathematical modelling to maximize production and improve efficiency for major oil & gas corporations. Led research on a new system that reduced build times by 40%.

Intermediate Software Engineer

2005 – 2008

BJ Pipeline Inspection Services Inc. (Baker Hughes)

C++ / C / OpenGL / PHP / SQL / Neural Networks / Borland / Windows / Linux

Visualised in-line device data with OpenGL graphics. Implemented neural network algorithms for anomaly detection. Led development of custom web-based GIS reporting software. Improved low-level programming algorithms for data processing.

Co-founder / App Developer / Instructor

2005 – 2008

SuRJE Software Solutions Inc. (self-employed, contract)

Objective-C / C++ / OpenGL / Chipmunk / PHP / JavaScript / Firebase / Google Maps / iOS / Linux

Created and published physics-based games for iOS. Handled technical writing, web development, Google Maps and Firebase integration and server administration. Developed course curriculum, instructed C++ programming and assessed student work with the Department of Continuing Education.

Software Researcher / Lead Developer

2005 – 2006

PlayStarMusic Corp.

C++ / PHP / Javascript / .NET / Windows / Linux

Led development and system administration for an iTunes-like music store. Automated DRM encoding and video streaming. Developed a prototype for a Google-competitive search engine.

Junior Software Engineer

2004 - 2005

Codefast.com

C++ / Qt / Wx / Shell script / Windows / Unix

Ported applications to operate under a new GUI library. Developed user interfaces for software lifecycle automation.

Web Developer / VR Artist

2002 - 2004

Nickle Arts Museum, University of Calgary (contract)

JavaScript / C++ / OpenGL / PHP / HTML / CSS / 3D Scanning / VR

Designed and developed an interactive online catalogue for the Gerald Hushlak art exhibit with JavaScript-based features. Contributed original generative art from graduate research. Collaborated on a pioneering VR installation using 3D scanning and CAVE immersion technology.

Research Assistant

2001 - 2004

VLab Computer Graphics Group / Artificial Intelligence Research Hub

C++ / OpenGL / PHP / MySQL / Qt / Solaris / Unix / Linux

Developed graphics software tools using fractal mathematics and biological modelling. Managed network setup and distributed computing infrastructure. Published five research papers and served as teaching assistant across three courses.

Education

- **Master's Degree** – University of Calgary 2002 – 2004 *Computer Science Thesis: Agent-based Development of Natural Transportation Networks*
- **Bachelor's Degree** – University of Calgary 1998 – 2001 *Computer Science*

Honours & Awards

- Departmental Nomination: Governor General's Gold Medal – Faculty of Graduate Studies 2005
- Postgraduate Scholarship NSERC D3 – Natural Sciences and Engineering Research Council of Canada 2005
- Graduate Student Scholarship – Alberta Learning 2004
- Best Student Paper Award (Co-author) – ACAL 2003
- CPSC Departmental Research Award – University of Calgary 2003

Portfolio

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Published Games

Ragin' Bull – iOS, Android (2020)
AR Picture Portal – iOS (2018)
Fragmented Reality – iOS (2018)
IceCream AR – iOS (2018)
Jewel Slide – iOS, Android (2017, 2019)
Word Slide & Find – iOS (2017)
A Word Theme – iOS, Android (2017, 2019)
CryptoQuip – iOS, Android (2011, 2019)
Rafter's Loot – iOS (2011)
Star Chase – iOS (2008)
Bitsy – iOS (2008)